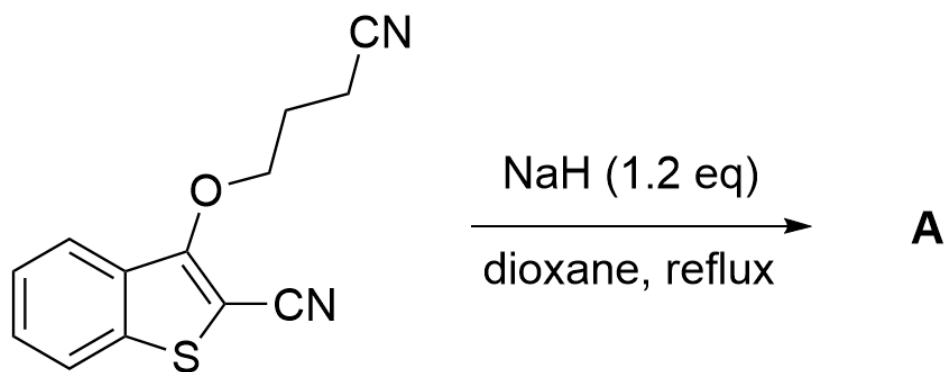


Question

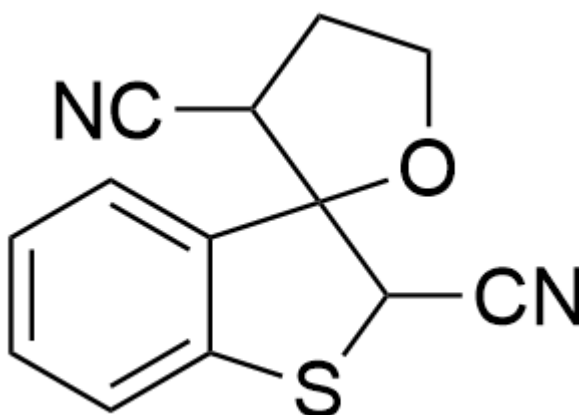


N#CC1=C(OCCCC#N)C2=CC=CC=C2S1>NaH(1.2equiv),dioxane,reflux>[A], **A** is the reaction product

Given that a new aromatic ring is formed in the reaction product **A**, provide the structural formula of **A**.

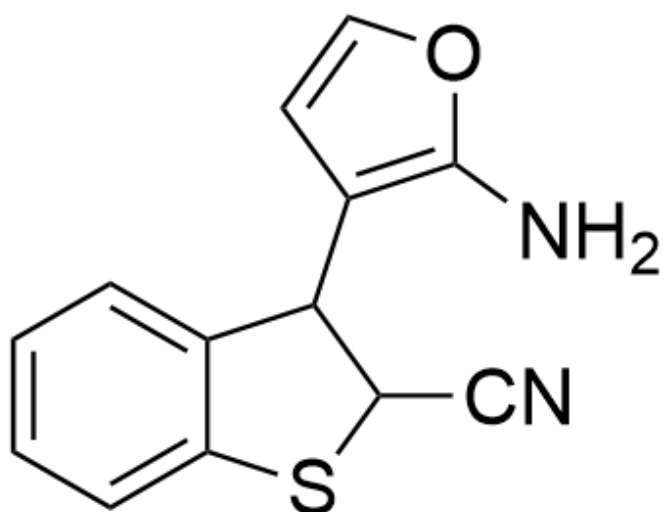
A. All other options are incorrect

B.



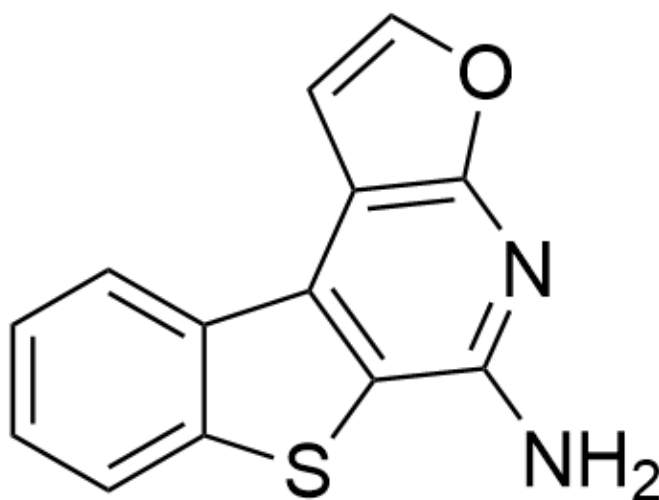
N#CC1SC2=CC=CC=C2C13OCCC3C#N

C.



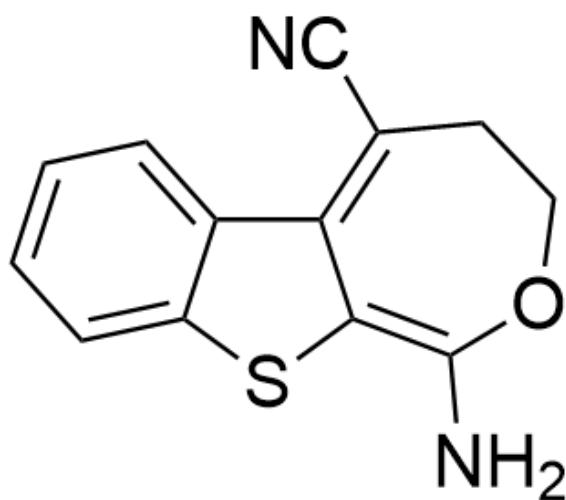
NC1=C(C2C(C#N)SC3=CC=CC=C32)C=CO1

D.



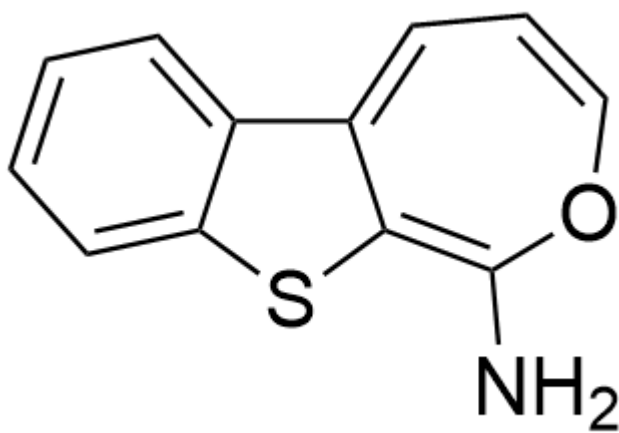
NC1=NC(OC=C2)=C2C3=C1SC4=CC=CC=C43

E.



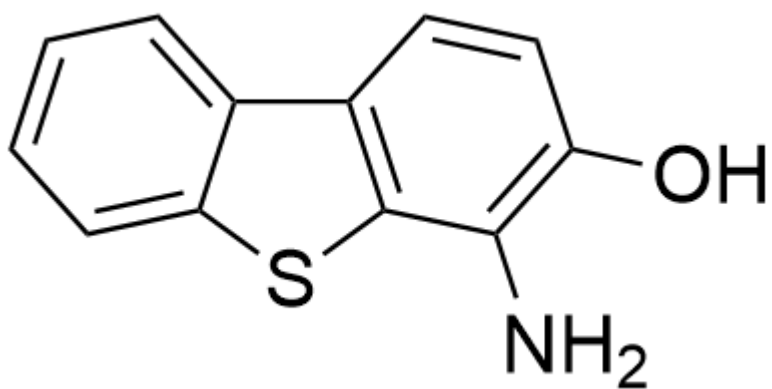
NC1=C2SC3=CC=CC=C3C2=C(C#N)CCO1

F.



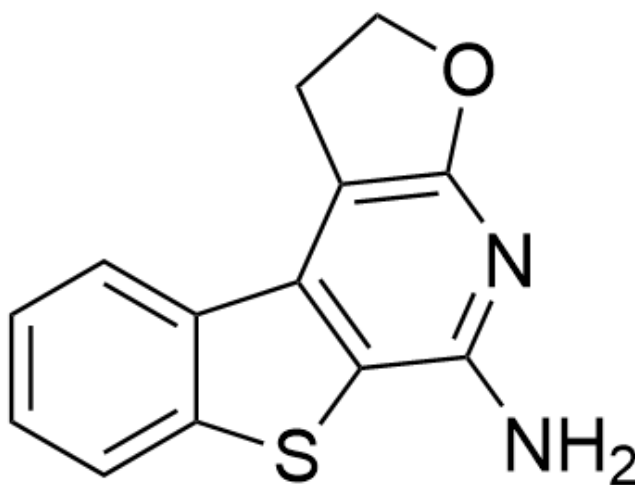
NC1=C2SC3=CC=CC=C3C2=CC=CO1

G.



OC(C=C1)=C(N)C2=C1C3=CC=CC=C3S2

H.



NC1=NC(OCC2)=C2C3=C1SC4=CC=CC=C43

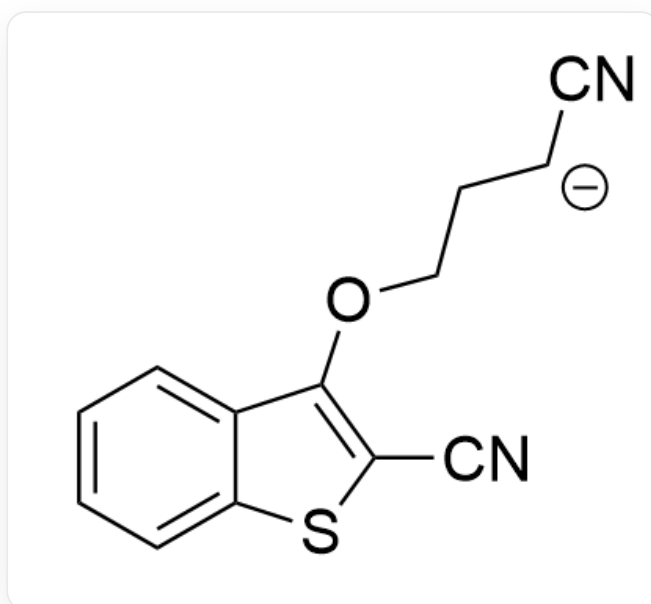
Answer

Correct Answer: **H**

Detailed Explanation

Based on the prompt, a new aromatic ring is formed in the reaction product **A**, excluding options B, E, and F.

First, the substrate is deprotonated by NaH at the position adjacent to the cyano group, yielding intermediate **1**.



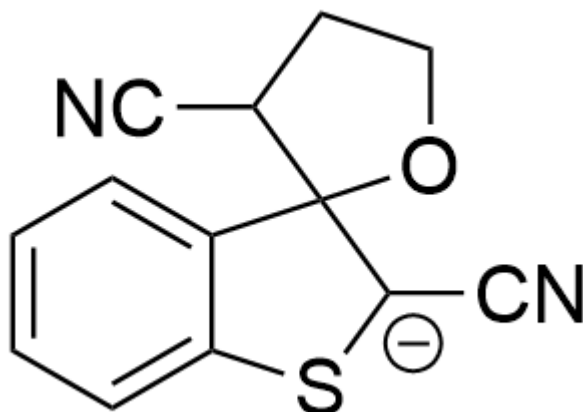
Intermediate **1**: N#CC1=C(OCC[CH-]C#N)C2=CC=CC=C2S1

CHECKPOINT

1 PTS

Intermediate **1**: N#CC1=C(OCC[CH-]C#N)C2=CC=CC=C2S1

Then, the formed carbanion undergoes Michael addition to the α - β unsaturated system to obtain intermediate **2**.



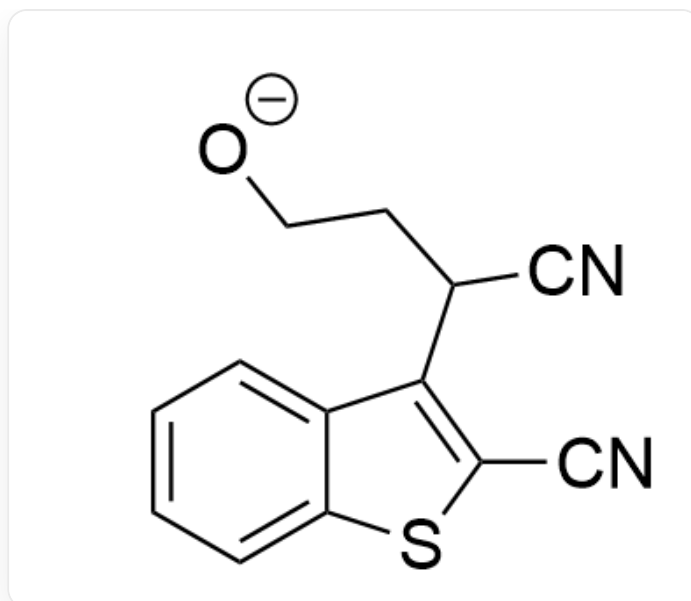
Intermediate **2**: N#C[C-]1SC2=CC=CC=C2C13OCCC3C#N

CHECKPOINT

1 PTS

Intermediate **2**: N#C[C-]1SC2=CC=CC=C2C13OCCC3C#N

Then, rearomatization occurs, and the oxyanion departs, yielding intermediate **3**.



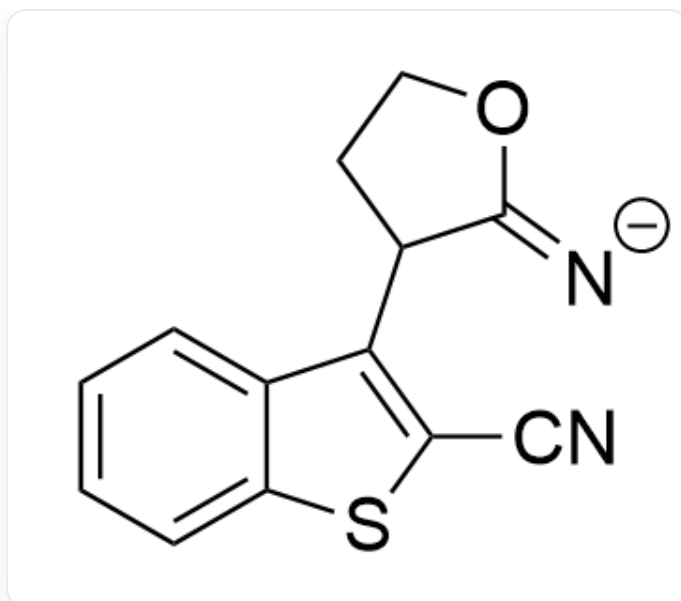
Intermediate **3**: [O-]CCC(C1=C(C#N)SC2=CC=CC=C21)C#N

CHECKPOINT

1 PTS

Intermediate **3**: [O-]CCC(C1=C(C#N)SC2=CC=CC=C21)C#N

The oxyanion attacks the cyano group to form a ring, yielding intermediate **4**.



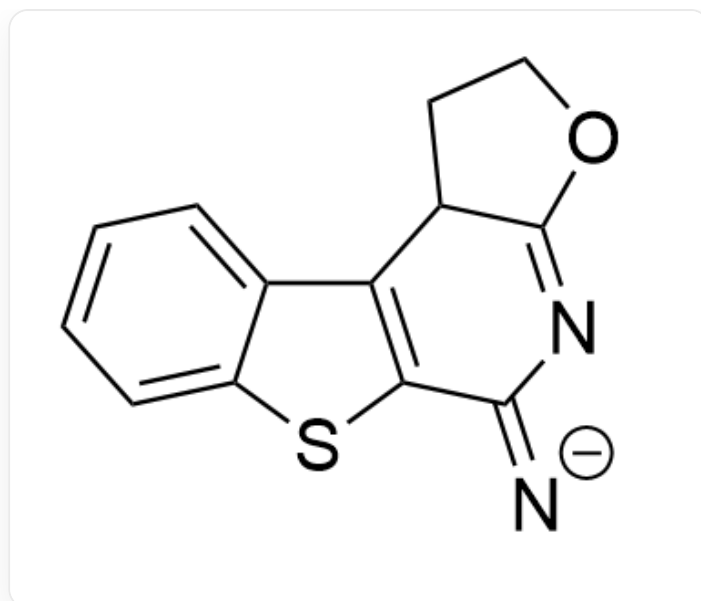
Intermediate 4: [N-]=C1C(C2=C(C#N)SC3=CC=CC=C32)CCO1

CHECKPOINT

1 PTS

Intermediate 4: [N-]=C1C(C2=C(C#N)SC3=CC=CC=C32)CCO1

The imine anion further attacks the cyano group to form a ring, yielding intermediate 5.



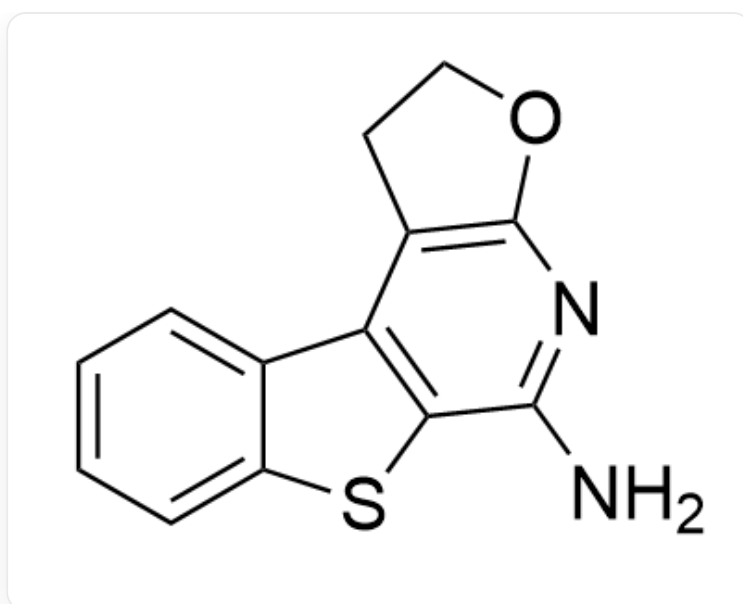
Intermediate **5**: [N-]=C1C2=C(C3CCOC3=N1)C4=CC=CC=C4S2

CHECKPOINT

1 PTS

Intermediate **5**: [N-]=C1C2=C(C3CCOC3=N1)C4=CC=CC=C4S2

Since the imine is unstable, and according to the prompt, a new aromatic ring is formed in the reaction product **A**, intermediate **5** further aromatizes to obtain product **A**.



Product **A**: NC1=NC(OCC2)=C2C3=C1SC4=CC=CC=C43